

SOIL REACTION:

An important property of soil solution is soil-reaction (ie whether the soil is acidic, neutral or alkaline). In soils where hydrogen ions ( $H^+$ ) predominates over hydroxyl ions ( $OH^-$ ), the reaction is said to be acidic. Where there are predominantly hydroxyl ( $OH^-$ ) ions, the reaction is said to be alkaline, where there are equal concentration of these two ions, the soil is said to be neutral.

SOIL ACIDITY: Soil acidity is the concentration of hydrogen ions ( $H^+$ ) in a solution is a higher - effect on soils. Acidity has a tremendous effect on soils.

Soil acidity is measured in pH units. Soil pH - is a measure of the concentration of hydrogen ions in the soil solution. The lower the pH of soil, the greater the acidity. pH is measured on a logarithmic scale from 1 - 14 with 7 being neutral.

CAUSES OF SOIL ACIDITY:

1. Application of Fertilizers like ammonium tetraoxosulphate II ( $NH_4)_2SO_4$  and ammonium trioxonitrate II ( $NH_4NO_3$ ).

Decomposition of organic matter (Humic acid produced). Leaching of nutrients.

oxidation of sulphur in the soil.

Acid rain especially in petroleum exploring areas.

- I.
- II.
- III.
- IV.
- V.

## Soil Alkalinity - Soil alkalinity occurs when -

There is a comparatively high degree of base saturation. The presence of salts, especially calcium, magnesium, and sodium carbonate, also gives a preponderance of hydroxyl ions over hydrogen ions in the soil solution. Under such conditions the soil is alkaline, some times very strongly, especially if sodium carbonate is present, a pH of 9 or 10 being common.

## SIGNIFICANCE OF SOIL PERTINENCE TO CROP PRODUCTION.

Soil pH affects many physical, chemical and biological reactions necessary for crop survival, growth and yield. Soil reaction affects crop production in the following ways:-

Nutrients availability - is strongly influence by pH - and is probably the second most important reason for soil pH management.

Generally, many nutrients including nitrogen (N), potassium (K), sulphur (S), calcium (Ca) and magnesium (Mg) are most available in slightly acidic to moderately alkaline soils while the availability of micro nutrients such as iron (Fe), manganese (Mn), boron (B), copper (Cu) and zinc (Zn) is increased in moderately acidic soils. Metal toxicity - Negatively affects root growth in acidic soils, and is the most important reason for managing soil pH. Certain metals such as Fe, Mn and Zn are released into the soil solution when pH becomes acidic.

These metals, particularly Al, damage plant roots by interfering with cell division and cell wall formation, and also interfere with the uptake, transport and use of nutrients, including Ca and P. Aluminium is not a plant nutrient, and is not actively taken up by crops, but can enter plant roots passively through osmosis. When it is present in the soil solution, Al<sup>3+</sup> and other forage legumes are especially sensitive to Al toxicity in acidic soils.

Efficacy and carry-over of soil-applied herbicides can also be affected by soil pH. In general, herbicides will be bound to the soil when herbicide and soil pH are similar. When herbicides are bound to the soil, weed control efficacy and herbicides degradation rate are both reduced. For example atrazine herbicides are acidic, and will be bound when applied to low pH soils. In such a situation, weed control in corn treated with atrazine would be reduced and the potential for injury to a following soybean crop increased. In this scenario, applying lime immediately prior to soybean would exacerbate the potential for crop injury, as soil bound herbicide is released from the soil in response to increase soil pH.

Nutrient transformation and nitrogen fixation are restricted in acidic soils due to reduced activity of the microorganisms that are responsible for these processes. Plant available forms of N, S and P are all reduced in acidic soils, as is symbiotic N fixation by leguminous crops.

iv.

iii.

## PROBLEMS ASSOCIATED WITH SOIL ACIDITY (PH)

Soil acidity affects plants in various ways viz:

Nutrients availability - plants absorb nutrients from the soil solution and in the right ionic form in which the plant can absorb them. Nutrients adsorbed on exchangeable sites on soil particles are in equilibrium with the soil solution and can also become available to plants. Nutrients which are a part of insoluble minerals are not available to plants. These nutrients are also called fixed nutrients.

The acidity of the soil influences the solubility of minerals and thus affect the availability of nutrients. Most plant nutrients are available at slightly acidic pH of 5.8 - 6.5.

Nutrients toxicity in acidic soils (metal) - In acidic soils with low pH level, metals such as Aluminium, Manganese might be released into the soil solution at high concentrations which may be toxic to many plants.

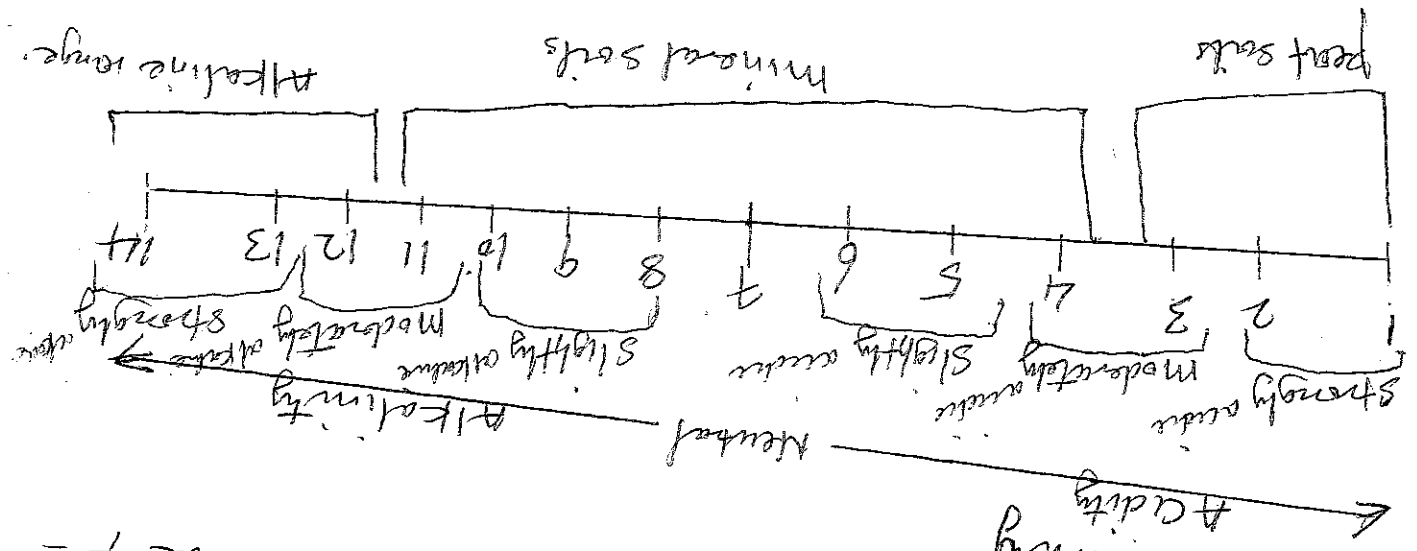
Effect on the activity of beneficial microorganisms - Some soil bacteria are responsible for many reactions in the soil, such as decomposition of organic matter (contributes nitrogen and phosphorus) and the nitrification process. These processes are significantly slowed down in acidic soils and therefore limit nitrogen and other nutrients availability.

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Soil pH: Soil pH is the negative logarithm of hydrogen ion concentration of soil solution and it is usually written as

$$pH = \log \frac{1}{[H^+]} \text{ or } pH = -\log [H^+]$$

In mineral soils, the pH range extends from nearly 3.5 — 10 or above. Some peat soils show pH of less than 3 and alkaline soils show pH exceeding 11. Humid region soils have the pH exceeding roughly from below 5 to above 7, and the arid region soils have pH ranging between 7 to approximately 9. The range of pH encountered worldwide is from 1—14, with 7 as the neutral reaction. pH of below 7 indicates acid and those above 7 means alkalinity



## THE EFFECT OF SOIL pH ON NUTRIENT UP-TAKE AND SOIL ORGANISMS.

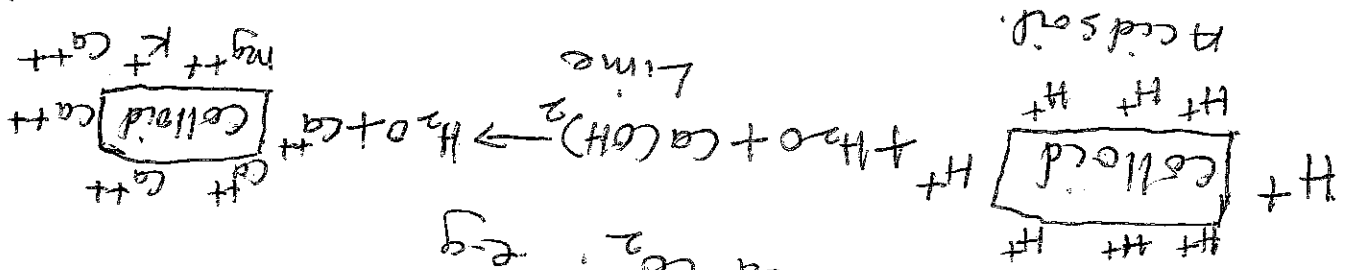
- I. It is a major factor in determining which trees, shrubs or grasses will dominate the land under natural conditions.
- II. pH influences the processes involved in the formation and development of soils.
- III. Most minerals are soluble in acid soils than in alkaline soils thus releasing ions toxic to plants e.g. Al.
- IV. It affects the availability of nutrients to the plants. Alkaline pH reduces the solubility of all the micronutrients (particularly Fe, Zn, Cu & Mn) except Mo and Cl. Within a pH range of 6.5 - 7.5, most of the essential nutrients (especially phosphates) are available to plants.
- V. The soil pH also affects plant growth by influencing the activities of beneficial soil microbes. Most N-fixing bacteria are not very active in strongly acidic soils. They release nitrogen and other nutrients for plant use - are depressed by strong acidity. Fungi usually tolerate acidity better than do other microbes.
- VI. Plant growth is also affected at high pH due to an excess of sodium ions both in soil exchange complex and solution, which actually deteriorate soils physical conditions for plant growth. Moreover, nutrient imbalance and sodium toxicity may also decrease plant growth.

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# Management of acid soil (acid soil can be managed through the following ways)

1. Addition of liming materials — soil acidity is the result of the accumulation of a predominance of  $H^+$  ions over  $OH^-$  ions. The bulk of  $H^+$  ions are held in close association with the colloidal complex. When lime is added to moist soil, the soil solution becomes charged with calcium ions.

These  $Ca^{++}$  ions exchange take place with hydrogen ions in the exchange complex. Hydrogen combines with  $OH^-$  ions to form neutral water or with  $CO_3$  to form unstable  $H_2CO_3$ , which is readily changed to  $H_2O$  and  $CO_2$ . e.g.



neutral soil  
Use of basic fertilizer — use of basic fertilizers like sodium nitrate, basic slag etc reduces the acidity in soils.

Use of rock phosphate — phosphate fertilization is almost essential for sustaining crop production in acid soils. As it is known that acid soils fix phosphorus in the soil. Application of rock phosphate in acid soil, soils provide an ideal situation for release of phosphorus (P) from rock phosphate (PP). The efficiency of rock-phosphate increases with increasing fineness.

At low pH (below 5.0) facilitates release of phosphate from rock phosphate and the effectiveness decreases - with increasing pH above 5.0. It is advisable to - apply rock phosphate several months ahead of liming to permit dissolution of rock phosphate at low pH. Rock phosphate is preferred source of phosphorus (P) in acid soils as it is a cheaper source and contains 30 to 50% CaO. CaO increases available calcium (Ca) in the soil, thus raises soil pH and lowers exchangeable Al content.

Soil management - proper soil and water management checks leaching of bases and enhances decomposition of organic matter.

Growing of acid tolerant crops - In acid soils, acid tolerant crops should be grown, choice of crops may be done according to soil pH. e.g. highly acid - tolerant crops are: rice, Sweet potato, oat, castor - Echinochloa, Paspalum etc.

Increasing the efficiency of nitrogen and potassium fertilizers in acid soils - The need for nitrogen - besides improving the efficiency of fertiliser system - to prevent N-fixing legumes in the cropping system - higher nitrogen (N) rates are recommended in high N fall areas to compensate for N losses. Split application of N definitely improves recovery.

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Crops that are tolerant to soil acidity are divided into the following groups:

Tolerant to Soil Acidity:

Red clover, Alfalfa, timothy, clover, corn, cucumber, beans, pepper, cauliflower.

Slightly acid tolerant crops are: Tomato, carrot, pea, etc.

Moderately acid tolerant crops are: Barley, wheat, maize, turnip, brinjal, cowpea, mung bean, pigeon pea, peanut.

Highly acid tolerant crops are: Rice, sweet potato, oat, castor, Echinochloa, Paspalum, etc.

Crops that are tolerant to soil acidity are divided into the following groups:

- Highly acid tolerant crops are: Rice, sweet potato, castor, Echinops.

The efficiency of potassium fertilizer can be increased by split application of  $K_2O$ . The split application is quite essential in acid soils having sandy texture. The top residues should not be removed from the field.

Water management - Acidification can be prevented by continuous flooding or saturation. Where that is not possible, the soils are drained and leached to remove the acid material. After submergence, the soil pH rises due to precipitation of aluminum hydroxide and by the reduction of ferric ion.

Crops like rice are tolerant to Al toxicity.

## Soil Amendment - Soil amendment is the

addition of materials to soil to improve its physical or chemical properties.

Its more municipal sewage sludge (SS) treatment districts turn to composting as a means of sludge stabilization and because of the rapid growth in the poultry industry, significant chicken manure (CM) and municipal SS generation will become available in increasing quantities.

Recycling wastes such as SS and CM for use as a low-cost organic fertilizer resulted in a positive effect on the growth and yield of a wide variety of crops and promoted the restoration ecology and economic functions of soil. The organic matter content of composted soil amendments is high, and its addition to agricultural soils often improves soil physical and chemical properties and enhances soil biological activities. Composts provide a stabilized form of organic matter that improves the physical properties of soils by increasing nutrient and water-holding capacity, total pore space, aggregate stability, erosion resistance, temperature insulation, and decreasing apparent soil density.

(11)

- Application of manures like compost, green and farmyard manures.
- Application of gypsum materials.
- Application of dolomite or calcite.
- Application of basic slag.
- Application of wood ash.
- Application of quick lime -  $\text{CaO}$ .
- Application of lime stone (Calcium carbonate -  $\text{CaCO}_3$ )
- Application of slaked lime -  $\text{Ca(OH)}_2$ .

Soil:  
The following are methods or materials for liming a well in this range.  
In most cases, soil fertility is improved by liming acid soil to 5.5 - 7.5. Most plants grow well in this range.

### LIMING MATERIALS/METHODS OF LIMING A SOIL

Lime is a calcium-containing inorganic mineral composed primarily of oxides and hydroxides. It is also the name for calcium oxide which occurs as a product of coal seam fires and altered lime stone. Usually calcium oxide and or calcium hydroxide. In most cases, soil fertility is improved by liming acid soil to 5.5 - 7.5. Most plants grow well in this range.

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## METHODS OF APPLICATION OF LIME.

To obtain quick action, lime is best applied to plowed land and worked into the soil as the seedbed is prepared. It should be mixed thoroughly with the surface half of the furrow slice. Top-dressing with lime and leaving it on the surface is seldom recommended except on permanent meadows and pastures.

However, it is often much more convenient and in the long run just as effective to apply the lime on the surface and plow it under.

## BUFFER AND BUFFERING CAPACITY:

A buffer is a compound that resists changes in pH when a limited amount of acid or base is added to it. The chemical composition of a buffer solution usually contains a weak acid or a weak base accompanied by its conjugate salt.

Buffering capacity can be defined as the measure of the efficiency of a buffer in resisting its change in pH. or buffer capacity may be defined as the quantity of a strong acid or strong base that must be added to one litre of a solution to change it by one pH unit. or buffer capacity quantifies the ability of a solution to resist changes in pH by either absorbing or desorbing  $H^+$  and  $OH^-$  ions.

# SOIL COLLOIDS

The colloid state refers to a two-phase system in which one material is dispersed through second phase, e.g. solid in liquid - clay in water (dispersion of clay in water). Liquid in gas - fog or clouds in atmosphere. The clay fraction of the soil contains particles less than  $0.002 \mu\text{m}$  in size. particles less than  $0.001 \text{ mm}$  size possess colloidal properties and are known as soil colloids.

## GENERAL PRINCIPLES OF SOIL COLLOIDS

1. Size - the most important - common property of inorganic and organic colloids is their extremely small size. They are too small to be seen with an ordinary light microscope. only with an electron microscope they can be seen. Most are smaller than  $2 \text{ micrometers}$  in diameter. Surface area - Because of their small size, all soil colloids expose a large external surface per unit mass. the external surface area of clay of colloidal clay is at least  $100 \text{ times}$  as well. These internal surfaces occur between plate-like crystal units that make up each particle and often greatly exceed the external surface area. The total surface area of soil colloids ranges from  $10 \text{ m}^2/\text{g}$  for clays with only external surfaces to more than  $800 \text{ m}^2/\text{g}$  for clays with extensive internal surfaces. The colloidal surfaces are in the upper  $15 \text{ cm}$  of a hectare of a clay soil could be as high as  $700,000 \text{ km}^2/\text{g}$ .
- 2.

3. Surface charges: Soil colloidal surfaces, both external and internal characteristically carry negative and/or positive charges. For most soil colloids, electro-negative charges predominate. Soil colloids both organic and inorganic when suspended in water, carry a negative electric charge. When an electric current is passed through a suspension of soil colloidal particles, they migrate to anode, the positive electrode indicating that they carry a negative charge. The magnitude of the charge is known as Zeta potential. The presence and intensity of the particle charge influence the attraction and repulsion of the particles towards each other, thereby influencing both physical and chemical properties.

Ionizable hydrogen ions.  
 Iso-morphism substitution.  
 Ionizable hydrogen ions - Ionizable hydrogen ions are hydrogen from clay layers the H and leaves an unneutralized negative charge on the oxygen.  $(-Al-O-O-Si-O)$ . The extent of ionized hydrogen ion depends on solution pH. More ionization occurs in more alkaline (basic) solution.

Iso-morphous substitution: The second source of charge on clay particles is due to the substitution of one ion for another of similar size and often with lower positive valency. In clay structures, certain ions fit into certain mineral lattice sites because of their convenient-size and charge. Dominantly, clays have  $Si^{+4}$  in tetrahedral sites and  $Al^{+3}$  in octahedral sites.

4. Adsorption of cations - As soil colloids possess negative charge they attract the ions of an opposite charge to the colloidal surfaces. They attract hundreds of positively charged cations such as  $H^+$ ,  $Mg^{2+}$ ,  $Ca^{2+}$  and  $Mg^{2+}$ . This gives rise to an ionic double layer.
5. Adsorption of water - In addition to the adsorbed molecules are associated with soil colloidal particles, some are attracted to the adsorbed cations; each of which is hydrated. These water molecules play a critical role in determining both the physical and chemical properties of soil.
6. Cohesion - Cohesion is the phenomenon of sticking together of colloidal particles that are of similar nature. This tendency of clay particles to stick together is primarily due to the attraction of the clay particles for the water molecules held between them.
7. Adhesion - Adhesion refers to the phenomenon of colloidal particles sticking to the surface of any other body or substance with which it comes in contact.
8. Swelling and shrinkage - Some clay (soil colloids) make deep cracks which at first allows rain to penetrate rapidly, later, because of swelling, such soil is likely to close up and become much more impervious than one dominated by kaolinite, chlorite, or fine grained micas. Vermiculite is intermediate in its swelling and shrinking characteristics.

Non-permeability — colloids as opposed to crystalline —  
are unable to pass through a semi-permeable membrane.  
Even though the colloidal particles are extremely small, they  
are bigger than molecules of crystalline dissolved in water.  
The membrane allows the passage of water and of the  
dissolved substance through its pores, but retains the colloidal  
particles.

8. Brownian Movement — When a suspension of colloidal  
particles is examined under a microscope, the particles  
seem to oscillate. The oscillation is due to the —  
collision of colloidal particles or molecules with those of  
the liquid in which they are suspended. So colloidal  
particles with those of water in which they are suspended  
are always in a constant state of motion. The smaller  
the particles, the more rapid is its movement.

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## SOIL ORGANIC MATTER

Soil organic matter represents an accumulation of partially decayed and partially synthesized plant and animal residues. Such material is continually being broken down as a result of the work of soil microorganisms. Consequently, it is a rather transitory soil constituent and must be renewed constantly by the addition of plant residues.

The organic matter content of a soil is small only about 3 to 5 percent by weight and a representative mineral topsoil.

## ORIGIN OF SOIL ORGANIC MATTER

All organic substances by definition contain carbon - the end product or ~~largest~~ proper part of organic matter.

However, the term soil organic matter covers all organic ~~in~~ components of a soil, these are:

1. living biomass (plant and animal tissues and microorganisms)
2. Dead roots and other recognizable plant residues.
3. largely amorphous and colloidal mixtures of complex organic substances no longer identifiable as tissues (leaf humus).

## FUNCTIONS OF ORGANIC MATTER:

i. improvement of soil mass structure.

ii. increased water holding-capacity of the soil.

iii. improved infiltration and erosion control.

iv. prevention of soil hardening.

v. improved soil cation exchange capacity.

vi. increased supply of slowly released inorganic nutrients.

vii. development of favourable environment for microbial activity in the soil.

viii. prevention of phosphate fixation by iron and aluminium oxides.

ix. Certain substances like gumme and benzogummes, which appear in the course of transformation of organic matter may play a specific physiological role and might increase the absorption capacity and length of roots.

x. Increase in the resistance of roots to some diseases.

xi. It improves the aeration of the soil.

## NATURE AND PROPERTIES OF HUMUS

Humus is defined as a complex and rather resistant mixture of brown or dark brown amorphous and colloidal substances modified from the original tissues or synthesized by the various soil organisms. It is a natural body and although it is exceedingly variable and heterogeneous, it possesses properties that distinguish it sharply from the original parent tissues and from the original humus developing during its synthesis.

Humus comprises of about 60 to 80% of the soil organic matter, having large molecules of variable structures and compositions.

Humus is the final products of decomposition of organic matter in soils. Some materials are very resistant to microbial attack in soils. Humus are classified as: humin, humic acid and fulvic acid.

## FACTORS AFFECTING THE NATURE AND

### QUANTITY OF ORGANIC MATTER IN SOILS.

Several factors affect the quantity of organic matter in soils, among which are:

1. The nature of soil-order (e.g. histosols which is basically an organically formed soil.

2. The influence and nature of climate.

3. The influence of natural vegetation.

4. The influence of drainage and soil texture.

5. The influence of agricultural management and tillage.

6. The influence of rotation, residues and plant nutrients.

### THE NATURE OF SOIL ORDER:

Though most soil orders contain considerable amounts of organic matter, yet they still differ in their level of composition and amounts. The nature of soil type, sources and formation influences the quantity and property of the soil and even the amounts of organic matter in the soils. The histosols - an deposit of organic matter has a very large amount and nature and origin of its formation.

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## THE INFLUENCE AND NATURE OF CLIMATE:

Climate plays an essential role in the quantity of organic matter in soils.

In nature, two important elements of climate are crucial in determining the amounts of organic-matter found in soils.

These elements are: Temperature and moisture. Soil temperature influences processes of organic-matter production (plant growth) and organic-matter destruction (microbial decomposition).

Similarly, soil moisture also exerts a major influence on the accumulation of organic-matter and nitrogen in soils. Under comparable conditions, the nitrogen and organic-matter content of soils increase as the effective moisture becomes greater.

## THE INFLUENCE OF NATURAL VEGETATION

Climate and Vegetation usually act together to influence the soil contents of organic carbon and nitrogen. The greater the plant population and number under a well-watered environment generally leads to greater addition of organic matter into soils.

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In forests and grass land vegetation, a relatively high ~~proportion~~ proportion of the plant residues consists of root matter which decomposes along side of grasses and leaves of plants into the soils. By this process, considerable amounts of organic matter are returned into the soil body.

### THE INFLUENCE OF TYPE & DRAINAGE

Soil texture and drainage are often responsible within a local landscape. Generally, soils high in silt and clay are higher in organic matter than sandy soils. The amount of organic residues returned to the soil is generally higher in fine textured soils and due to formation of clay humus complexes that protects the organic matter from degradation or loss.

poorly drained soils generally accumulate and nitrogen than in similar soils under aeration still on poorly drained environment, sufficient organic matter may accumulate to form

# THE INFLUENCE OF AGRICULTURE

## MANAGEMENT AND TILLAGE PRACTICES:

Interests thereby enhancing organic matter levels in soils.

Generally cultivated lands contains much lower levels of both soil nitrogen and organic matter than the comparable areas under natural vegetation. All organic matter returned into the soil during agricultural management and practices. Also soil tillage facilitates to increase decomposition. More accessible in soil organic content when a rapid decline occurs under cultivation. Smaller decline cleared. Today, both conventional and modern conservation tillage practices are vital in restoring organic matter into soils. Practices such as stubble-mulch, zero or no tillage systems etc. influences the amount of organic matter of organic matter at a given soil.

## Sources of organic manure

There are three types of organic manure. Vg:-

- a. Green manure (b) Farm-yard manure (c) Compost manure.
9. Green manure — is formed from leguminous crops and other fresh plants which are ploughed into the soil while they are still growing. The plants have to be stage so that they are very young, i.e. before they are ploughed to the soil. e.g. legumes like mung, cowpea, cantharid, *calopogonium*, *pisum* and grasses.
6. Farm-yard manure — Farm-yard manure is a combination of animal wastes such as animal dung or faeces, series of decomposition, before the manure is applied to the soil. Putting <sup>and</sup> generally produce manure of a higher nutritive value than larger animals. Large and sheep also produce manure which are rich in its decomposition before it can be applied to the soil, otherwise the heat produced during decomposition would burn the roots of plants.
- c. Compost manure — Compost manure is the type of manure formed as a result of the rotting down of plants and animal remains in heaps or — is applied to the soil.



# PLANT'S NUTRIENTS AND THEIR FORMS OF AVAILABILITY:

Plants require certain food substances in form of mineral salts or nutrients for their growth and development. There are about fourteen mineral-salts required by plants. They are divided into two groups, that is macro nutrients or minerals and micro nutrients or minerals.

## MACRO NUTRIENTS:

These are the minerals salt required in large amount by plants and their deficiency can easily manifest on the plants. The macro nutrient are - Nitrogen, Phosphorus, Potassium, Calcium, Magnesium and Sulphur. Plants growth may be retarded because these elements are actually lacking in the soil, because they are not adequately balanced by other nutrients because they are not adequately balanced by -

These are the minerals salts required in small quantity by higher plants thereby justifying the name micro nutrient or trace elements. Such a designation does not mean that they are less essential than the macro-nutrients but merely that they are ~~needed~~ needed in small quantities. The main nutrients are: Iron, Manganese, Copper, Zinc, Boron, Molybdenum, Chlorine and Cobalt.

### MICRO NUTRIENTS:

The main nutrients are: Nitrogen, Phosphorus, Potassium, Calcium, Magnesium and Sulfur. Plants growth may be retarded because these elements are actually lacking in the soil, because they ~~are~~ become available for study or because they are not adequately balanced by other nutrients.

The essential nutrients and forms in which plants absorb them in the soil are as follows:

### Macro nutrient

Nitrogen —  $\text{NO}_3^-$ ,  $\text{NH}_4^+$   
 Phosphorus —  $\text{H}_2\text{PO}_4^-$ ,  $\text{HPO}_4^{2-}$   
 Potassium —  $\text{K}^+$   
 Calcium —  $\text{Ca}^{++}$   
 Magnesium —  $\text{Mg}^{++}$   
 Sulphur —  $\text{SO}_4^{2-}$ ,  $\text{SO}_3^{2-}$

### Micro nutrients

Iron —  $\text{Fe}^{++}$   
 Manganese —  $\text{Mn}^{+2}$   
 Copper —  $\text{Cu}^{++}$   
 Zinc —  $\text{Zn}^{++}$   
 Boron —  $\text{B}_4\text{O}^{2-}$   
 Molybdenum —  $\text{MoO}_4^{2-}$   
 Chlorine —  $\text{Cl}^-$   
 Cobalt —  $\text{Co}^{+++}$

## FUNCTIONS AND DEFICIENCY SYMPTOMS OF PLANT NUTRIENTS:

### NITROGEN:

Nitrogen is an inert, odourless, tasteless gas is unavailable to plants unless it is "fixed" in the soil in soluble inorganic forms. Plants absorb nitrogen from the soil mainly in forms of  $\text{NO}_3^-$  and  $\text{NH}_4^+$  ions.

### FUNCTIONS OF NITROGEN IN PLANTS:

It is essential for plant growth. Necessary for reproduction.

Found in all proteins, enzymes, hormones and protein of all living cells.

Essential for the increase in the yield and quality of grains.

It also increases the number of leaves to not develop

It increases carbohydrate synthesis.

It promotes vegetative growth.

Makes the leaf deep green in colour thereby ~~encouraging~~ encouraging photosynthesis.

It also promotes the uptake of potassium and phosphorus by plants.

### DEFICIENCY OF NITROGEN IN PLANTS:

Stunted growth.

Older leaves turn yellow or have chlorosis.

The leaves tend to drop.

PHOSPHORUS: In world agriculture, phosphorus is second to nitrogen in importance for production of healthy plants and profitable yields.

FUNCTIONS OF PHOSPHORUS (P)

Essential for growth and development.  
It is essential for respiration, fat and albumen formation.  
Necessary for cell division because it is a constituent of nucleoprotein.

Necessary for carbohydrate synthesis and degradation.

Essential for seed germination.  
Necessary for seed and fruit formation.

Also very essential for flowering and root development.  
It hastens fruit ripening thus removing the stress in nitrogen effects.

It increases crop resistance to diseases and prevents lodging in cereals.

Improves the palatability of forages and vegetables.

DEFICIENCY OF PHOSPHORUS:

Leaves turn to purple or green colour.  
Stunting of the green part of a plant.  
Reduction in metabolism in certain organs of the plants.  
Inhibition of flowering, fruits and seed formation.

POTASSIUM (K): Potassium is the third most essential after nitrogen and phosphorus.

### FUNCTIONS OF POTASSIUM:

Essential for plant metabolism.  
Helps in carbohydrate formation.  
Necessary for starch translocation.  
It is required for the neutralisation of organic acids in plants.

Helps in nitrate uptake from soil.  
It strengthens the stem and stalk thereby preventing lodging in the plants.

It activates various plant enzymes.

It helps in the opening of stomata, thereby affecting the amount of water and air in the leaves of plants.

It is a major constituent of plant tissues.

### DEFICIENCY OF POTASSIUM:

Weak and slender stem.  
Retarded and delayed growth.  
Premature leaf fall.  
Lodging in plants.

## FUNCTIONS OF CALCIUM (Ca)

Helps in high flocculation is good aeration and water infiltration and retention.  
It improves the pH of the soil so that nitrogen fixation can be carried out.  
It strengthens plant cell walls.

It controls toxicity of aluminium, calcium and sodium.  
It is necessary for the normal growth of root tips.  
It is the most abundant element in plant leaves.

## DEFICIENCY OF CALCIUM:

It causes stunting of the root system.  
It causes acidity.

Weak slender plants.  
Pale yellow colour of the leaves.

## FUNCTIONS OF MAGNESIUM (Mg)

Essential constituent of chlorophyll which is necessary for photosynthesis.  
Helps in the transportation of phosphate ions, hence it is important in oil crops.  
It activates several enzymes.

## DEFICIENCY OF MAGNESIUM:

Chlorosis or yellowing of leaves along the leaf veins.  
Stunted growth.  
Photosynthesis cannot take place.

## FUNCTIONS OF SULPHUR(S).

It is a member of many plant proteins eg cysteine and methionine.  
It is also found in plant hormones like biotin and thiamine.

It is necessary for chlorophyll formation.  
It is also required for carbohydrate metabolism and nitrogen fixation by legumes.

## DEFICIENCY OF SULPHUR.

Yellowing of leaves or chlorosis.  
Stunted growth.  
Disturbed photosynthesis.

## FUNCTIONS OF THE MICRONUTRIENTS:

### FUNCTIONS OF IRON (Fe).

It is necessary for chlorophyll formation.  
Important in enzyme systems for oxidation-reduction reactions.  
It is essential for the synthesis of proteins contained in the chloroplasts.

### DEFICIENCY OF IRON:

Chlorotic condition in leaves which become pale green.